

DECISION BRIEF

# Clinical Document Generation v2

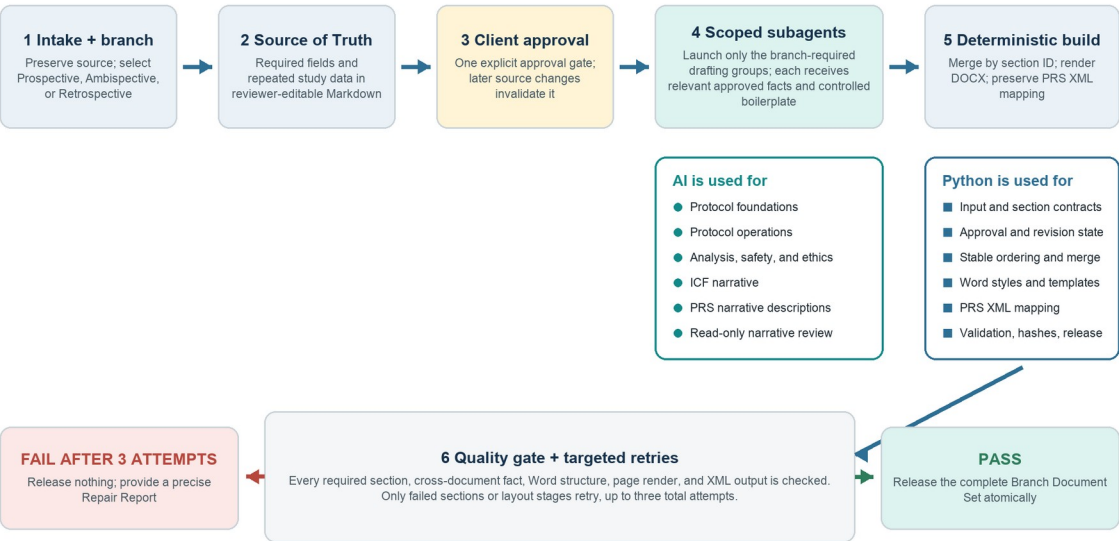
A controlled rebuild for complete, consistent, Word-ready clinical documents

**RECOMMENDATION** Approve a clean rebuild around one public workflow and six production Python modules. AI remains responsible for scoped clinical narrative; deterministic code controls facts, formatting, validation, and release.

## How the proposed skill works

### Clinical Document Generation v2

Human-approved facts. Scoped AI drafting. Deterministic Word delivery.



Why this split works: AI handles context-sensitive writing; deterministic code handles repeatability, formatting, traceability, and release control.

No partial delivery: the complete branch package passes together, or the client receives a Repair Report.

## Where subagents are used - and where they are not

The design deliberately uses AI only where clinical language requires judgment. Everything that must be repeatable, auditable, and formatting-stable remains deterministic.

| Decision area    | Subagents                                                                               | Deterministic Python                                                                                        |
|------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| When             | After the Source of Truth is explicitly approved.                                       | From intake through approval, and again for assembly, QA, and delivery.                                     |
| What             | Draft narrative sections and perform read-only narrative or visual review.              | Validate inputs, enforce contracts, merge sections, render DOCX, map PRS XML, record evidence, and release. |
| Inputs           | Only the approved fields relevant to assigned section IDs, plus controlled boilerplate. | The full approved source, versioned contracts, templates, and accepted drafts.                              |
| Authority        | Propose text and findings. They cannot approve facts or release files.                  | Owns pass/fail decisions, retry targets, artifact hashes, and atomic handoff.                               |
| Failure behavior | Retry only failed sections, up to three total attempts.                                 | Block release after the retry limit and produce a classified Repair Report.                                 |

### Five available drafting groups - only the branch-required groups run

- Protocol foundations - rationale, objectives, population, and study design.
- Protocol operations - procedures, visits, measurements, and schedules.
- Protocol analysis and oversight - endpoints, statistics, safety, ethics, and risks.
- ICF narrative - client-template language adapted only with approved study facts.
- PRS narrative - brief and detailed descriptions while preserving the existing XML mapping.

*Prospective and Ambispective studies use all five groups. Retrospective studies use the three Protocol groups and do not launch ICF or PRS drafting.*

### Why this division makes sense

- **Better writing.** Specialized agents can focus on one clinical concern instead of juggling the whole package.
- **Lower invention risk.** Each agent sees only relevant approved facts and controlled clinical boilerplate, not raw or unrelated material.
- **Faster, safer retries.** A failed section is redrafted in isolation; accepted sections are reused rather than rewritten.
- **Stable output.** Code, not AI, controls section order, Word styles, tables, fields, headers, footers, XML, and file release.

**KEY CONTROL** Subagents may select, adapt, and connect approved boilerplate, but they may not invent study-specific procedures, risks, benefits, safety obligations, or regulatory claims.

## A manageable six-module production architecture

| Production module | Single responsibility                                                                       |
|-------------------|---------------------------------------------------------------------------------------------|
| workflow.py       | Only public endpoint: prepare, approve, validate, and generate.                             |
| contracts.py      | Branch input requirements and document section contracts.                                   |
| drafting.py       | Subagent assignments, structured section drafts, targeted retries, and deterministic merge. |
| rendering.py      | Protocol and ICF DOCX generation using contracted client templates.                         |
| quality.py        | Content, cross-document, structure, renderer, visual, evidence, and release gates.          |
| prs_xml.py        | Preserved PRS XML mapping and validation behavior.                                          |

### Branch deliverables

| Study branch  | Required client package            |
|---------------|------------------------------------|
| Prospective   | Protocol DOCX + ICF DOCX + PRS XML |
| Ambispective  | Protocol DOCX + ICF DOCX + PRS XML |
| Retrospective | Protocol DOCX                      |

### Word compatibility without requiring Word on every machine

- **Output standard.** The skill generates standards-compliant DOCX files designed for Microsoft Word.
- **Active renderer.** For visual QA it uses Microsoft Word when available, then LibreOffice, then Apple Pages, and records which renderer produced the evidence.
- **Honest certification.** If no renderer exists, structural checks may run, but the package is not reported as visually passed.
- **Template guarantee.** Only bundled, contracted templates are supported until a new template completes onboarding and acceptance testing.

**PORTABILITY** The client can open and update the final DOCX in Microsoft Word even when the development machine used LibreOffice or Pages for visual QA.

## Release controls and approval criteria

| Current risk                 | New control                                                                                            | Evidence before release                                                         |
|------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Empty or weak sections       | Versioned section contract checks evidence and substance, not just non-empty text.                     | Every required leaf section passes; container headings may group children.      |
| Unsupported clinical details | Agents receive approved facts plus controlled boilerplate; cross-document checks compare shared facts. | No unsupported study-specific claim or contradiction.                           |
| Formatting quirks            | Word-native styles, fields, tables, headers, footers, and per-page visual inspection.                  | No clipping, overlap, blank pages, orphan headings, split rows, or style drift. |
| False-positive delivery      | Atomic branch release tied to artifact hashes and renderer evidence.                                   | Every required file passes together; later mutation invalidates certification.  |
| Unmaintainable scripts       | One public workflow and six focused production modules.                                                | Acceptance tests exercise the full workflow rather than private script chains.  |

### Acceptance corpus

- Sparse-complete and rich-complete studies for each of the three branches (at least six canonical studies).
- Current failing Prospective and Ambispective cases converted into regression tests, plus a complete Retrospective end-to-end case.
- Whole-workflow tests using real bundled templates, accepted section drafts, final rendering, quality gates, and atomic handoff.
- Golden regression protection for the existing PRS XML output and full-corpus reruns after contract or boilerplate changes.

**APPROVAL REQUESTED** Authorize the clean rebuild and replacement of the current script chain, subject to the six-module boundary, the human Source-of-Truth approval gate, scoped subagent use, three-attempt retry policy, preserved PRS XML behavior, and the release criteria above.

**Definition of done:** with all obligatory inputs complete, every required section is substantive, every branch deliverable passes its contract, and the complete package renders cleanly before client release.